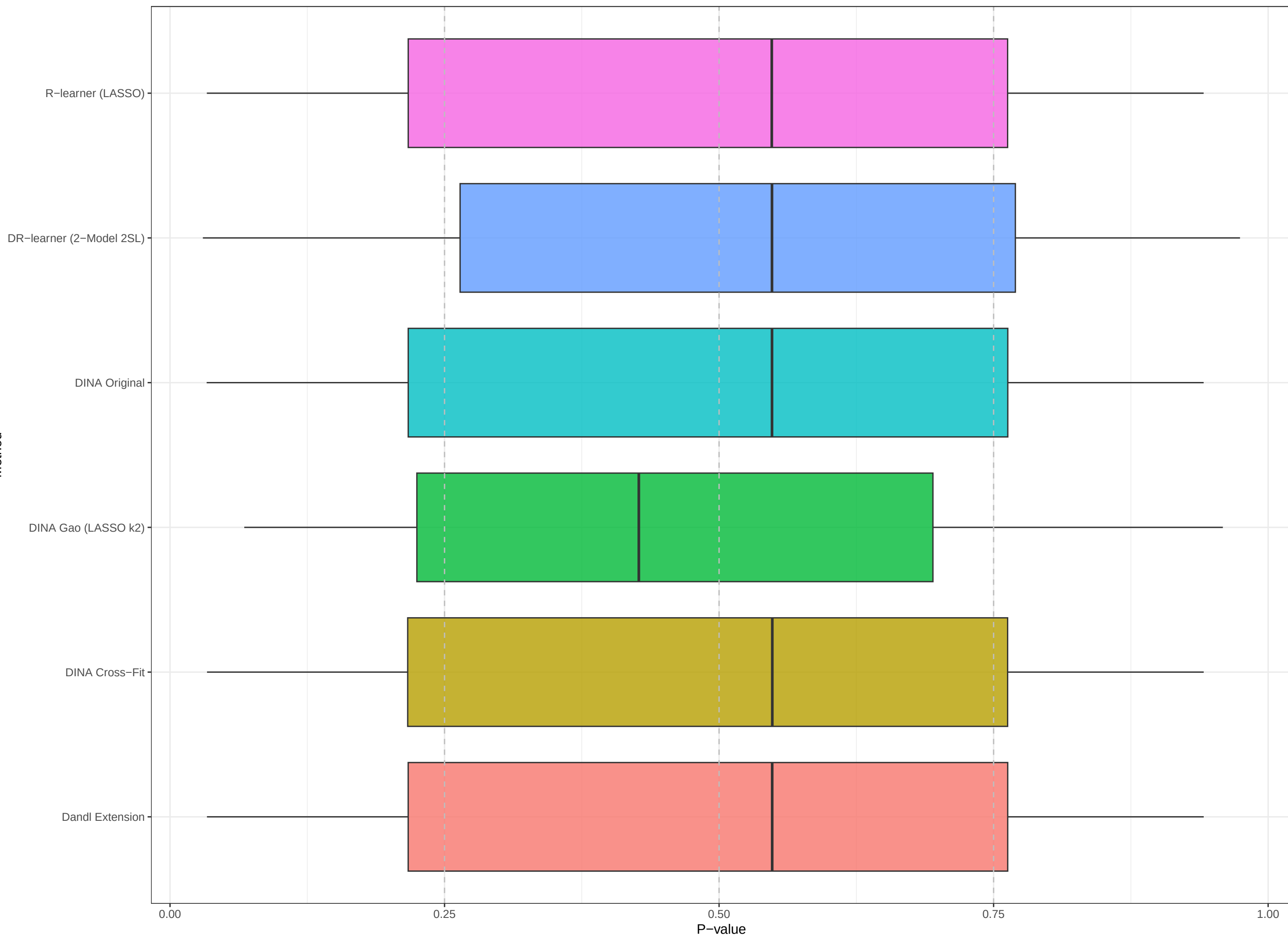
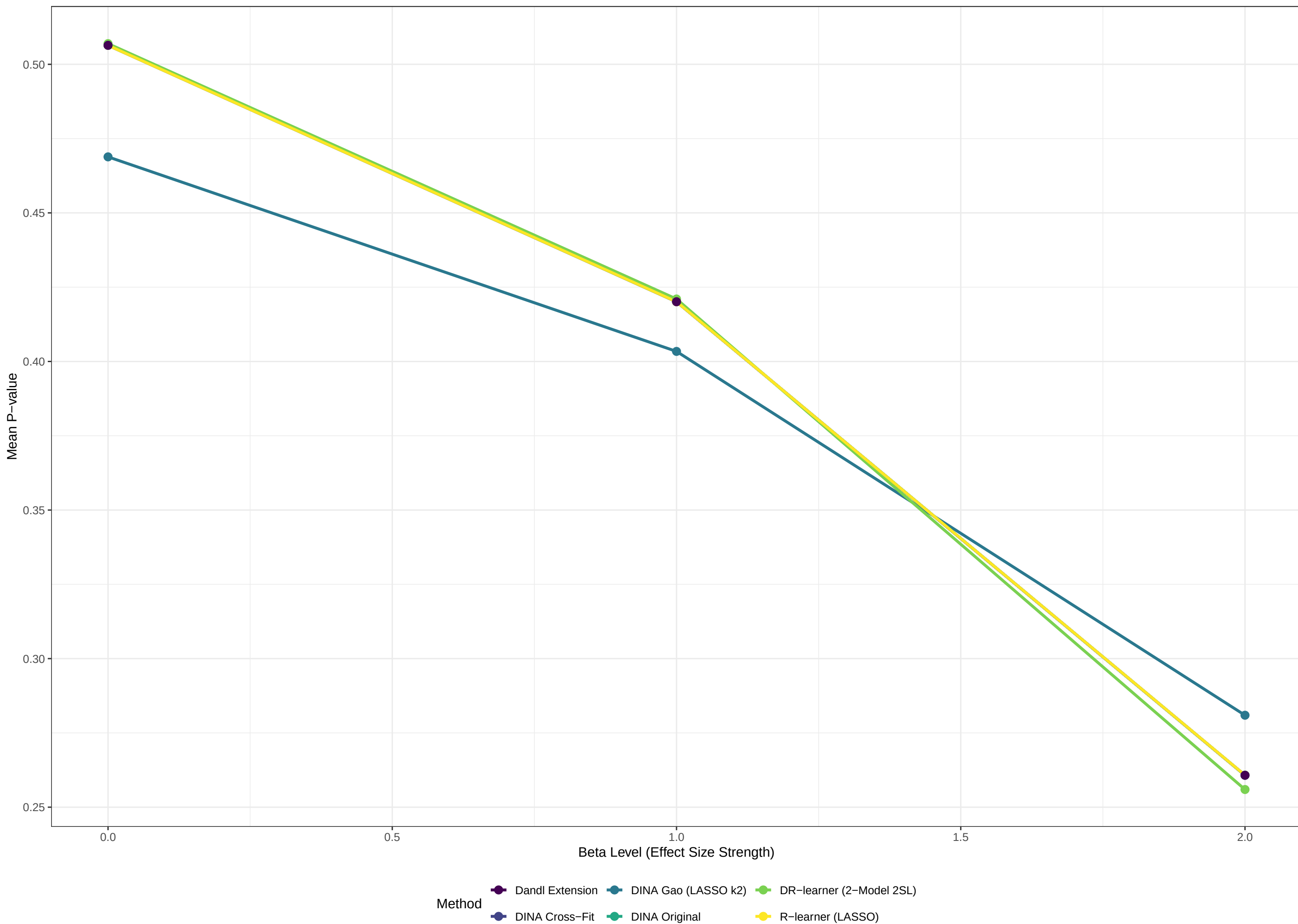


Objective 1(i): P-value Under Null (Scenario 4, $\beta=0$)
P-values should be $\sim \text{Uniform}[0,1]$ when no heterogeneity exists

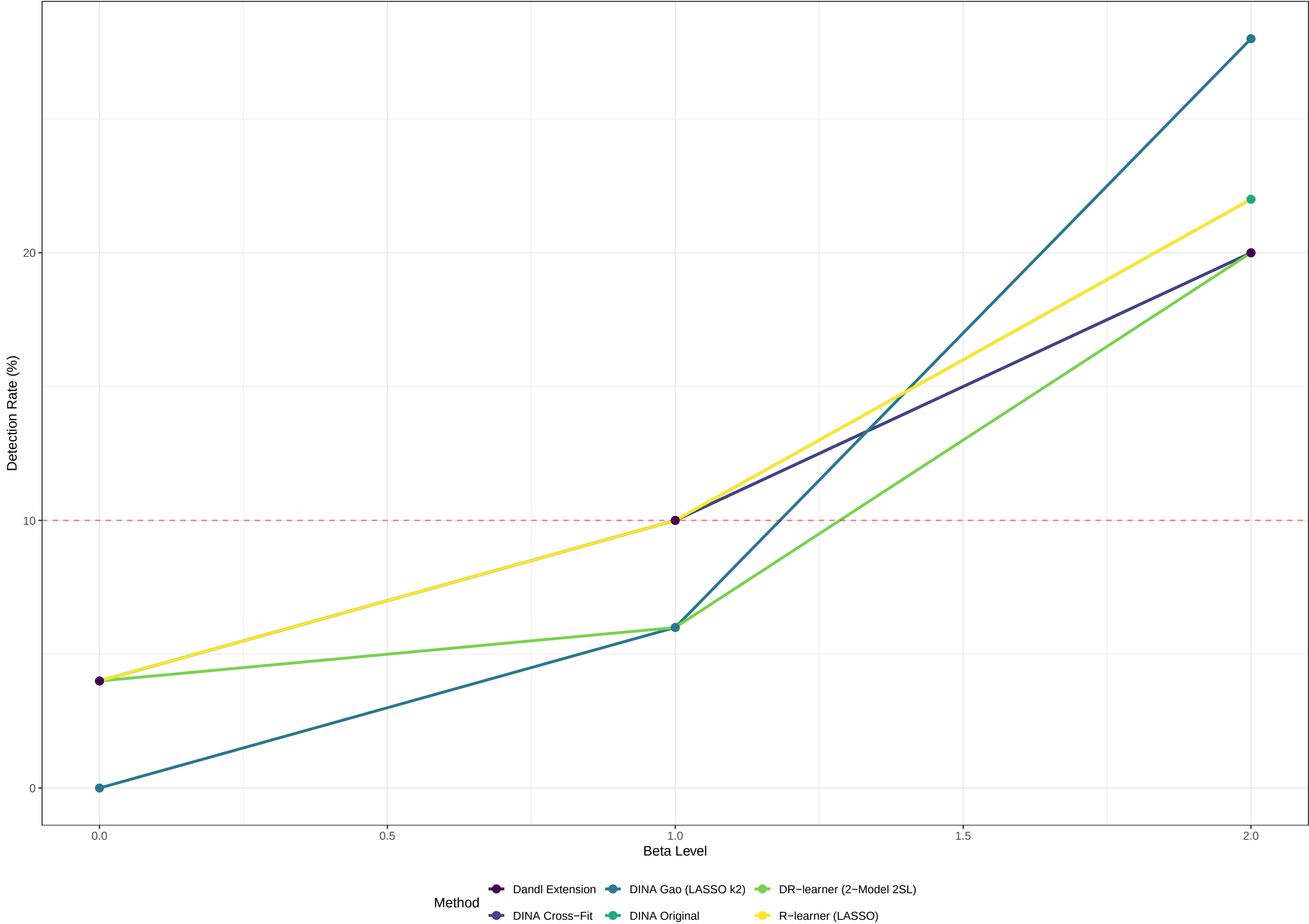


Objective 1(ii): Power Analysis (Scenario 4)

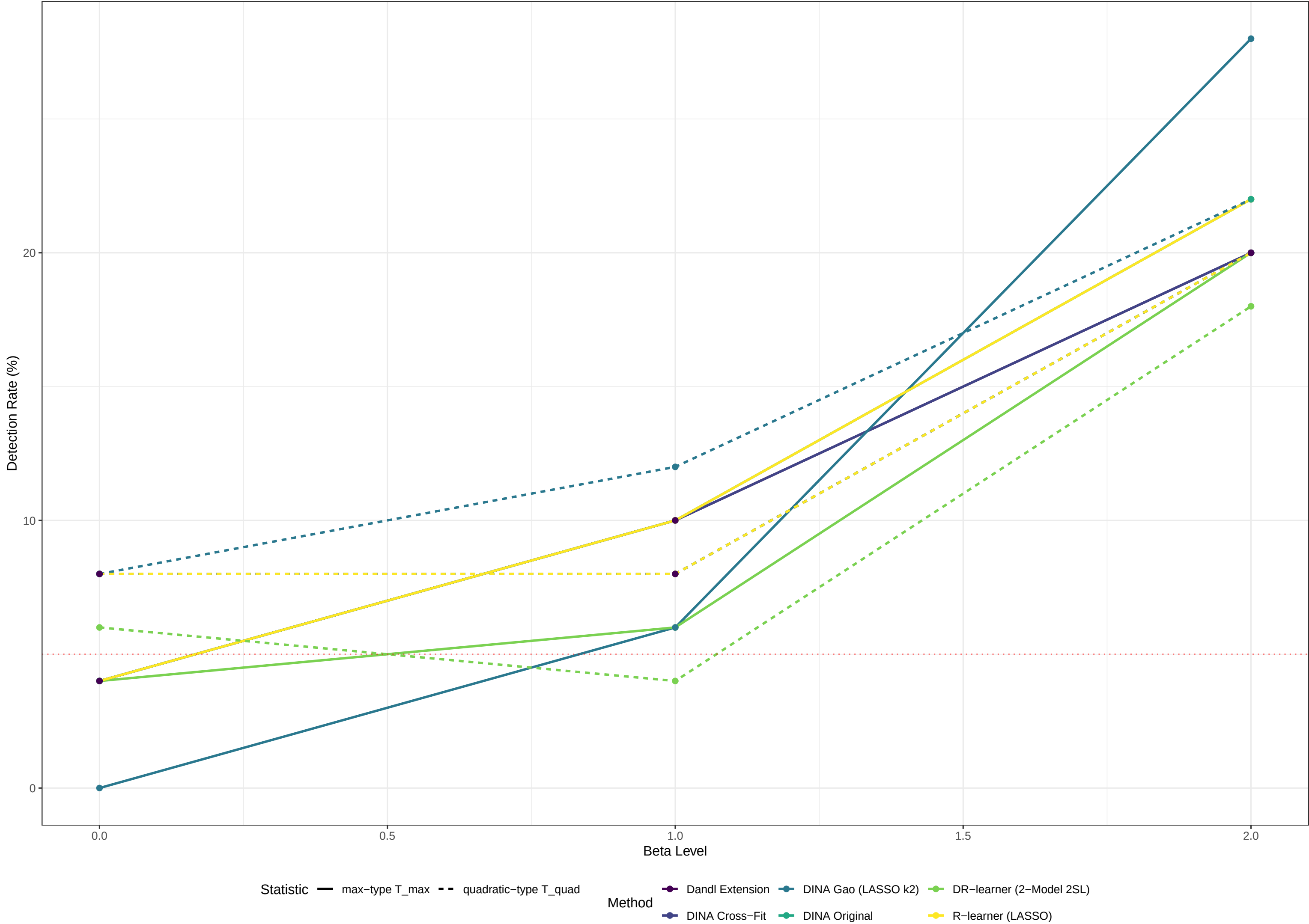
Lower mean p-value indicates better power to detect heterogeneity



Objective 1: Detection Rate (Scenario 4)
% of sims detecting heterogeneity (p < 0.05). Red line = nominal ..

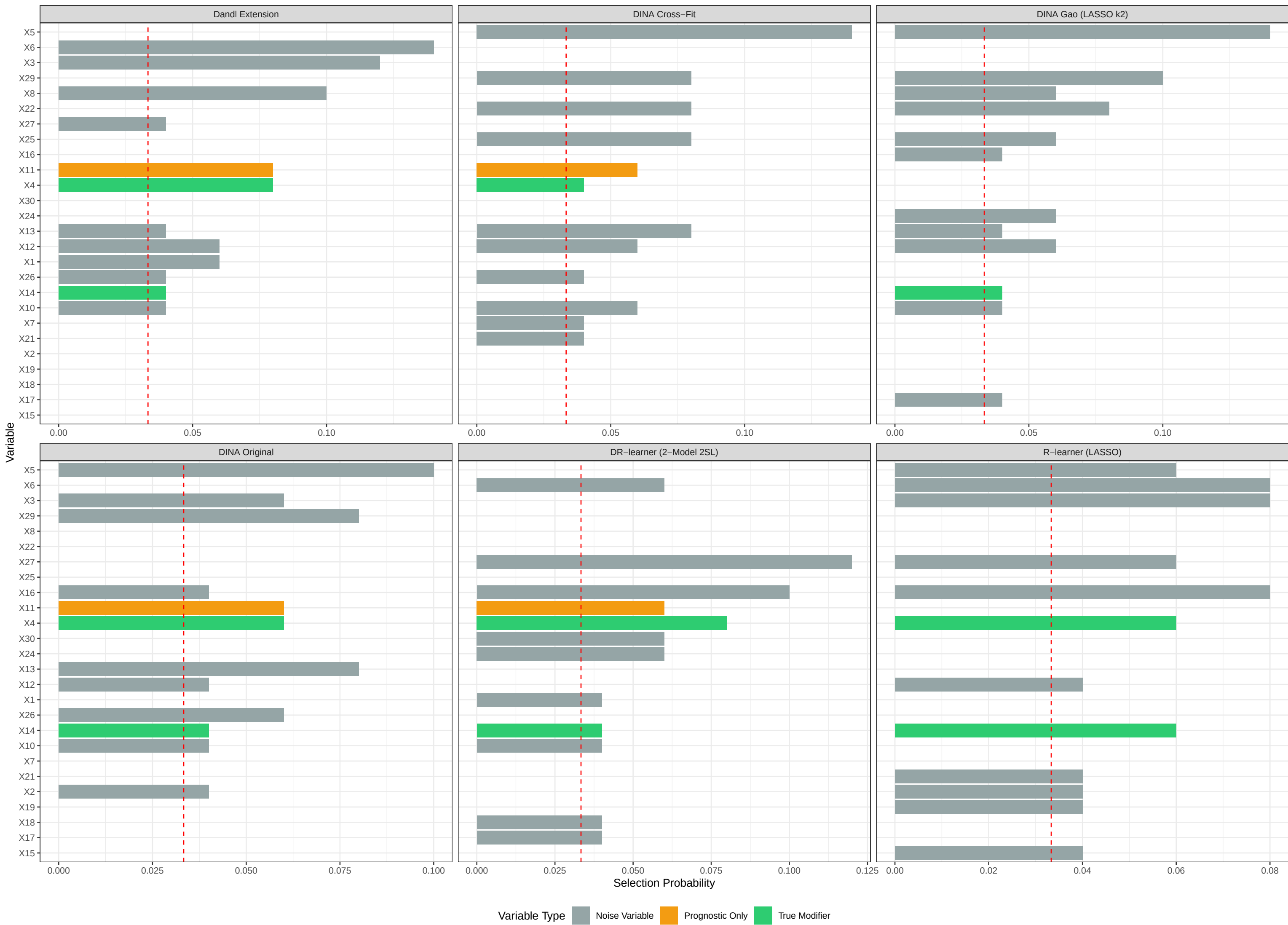


Objective 1: Max- vs Quadratic-type Power (Scenario 4)
Permutation test on each method's pseudo-outcome. Red dotted = nominal $\alpha = 5\%$.



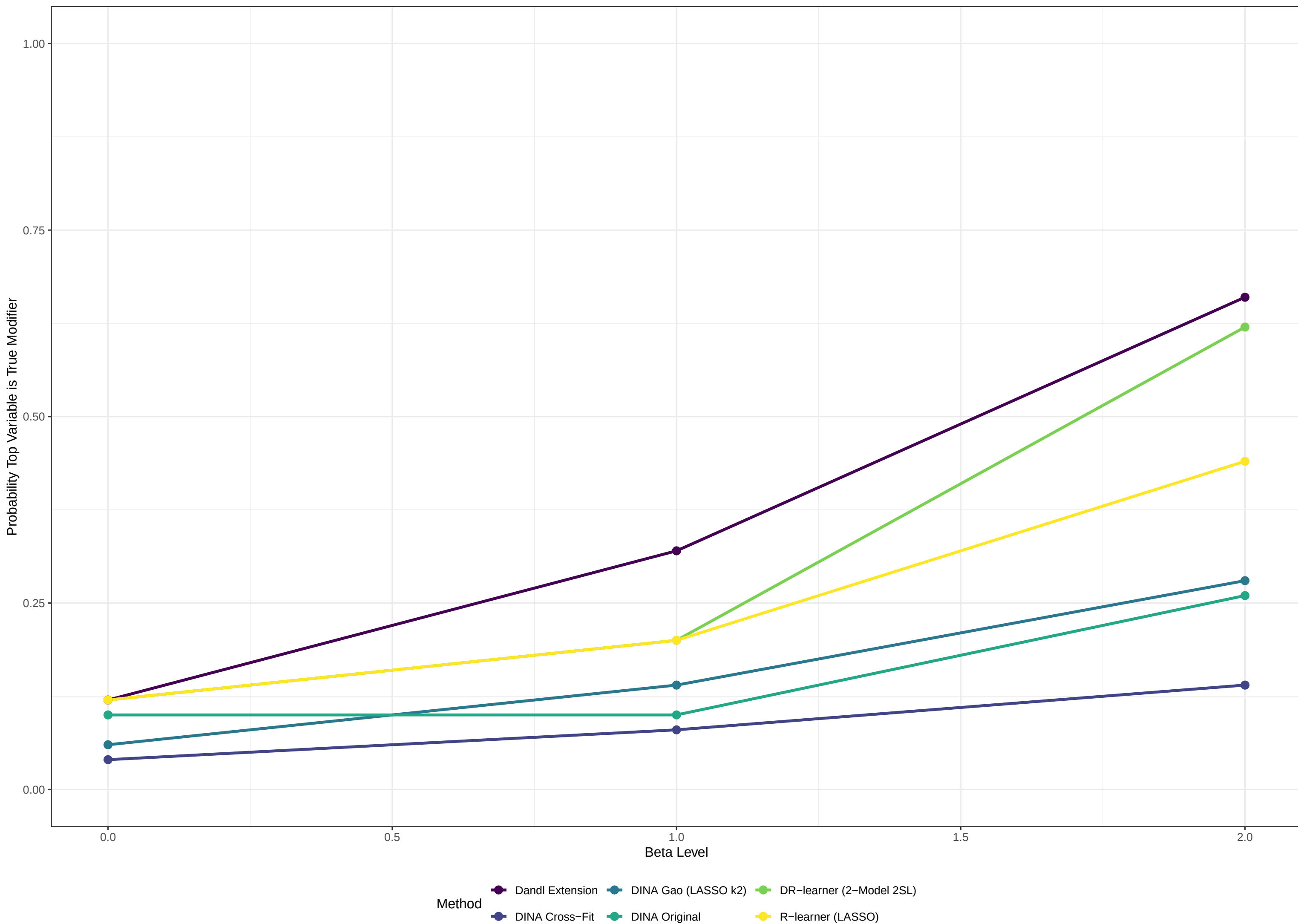
Objective 2(i): Top Modifier Under Null (Scenario 4, $\lambda=0$)

Unbiased methods select all vars equally (~3.3% for 30 vars, red line)



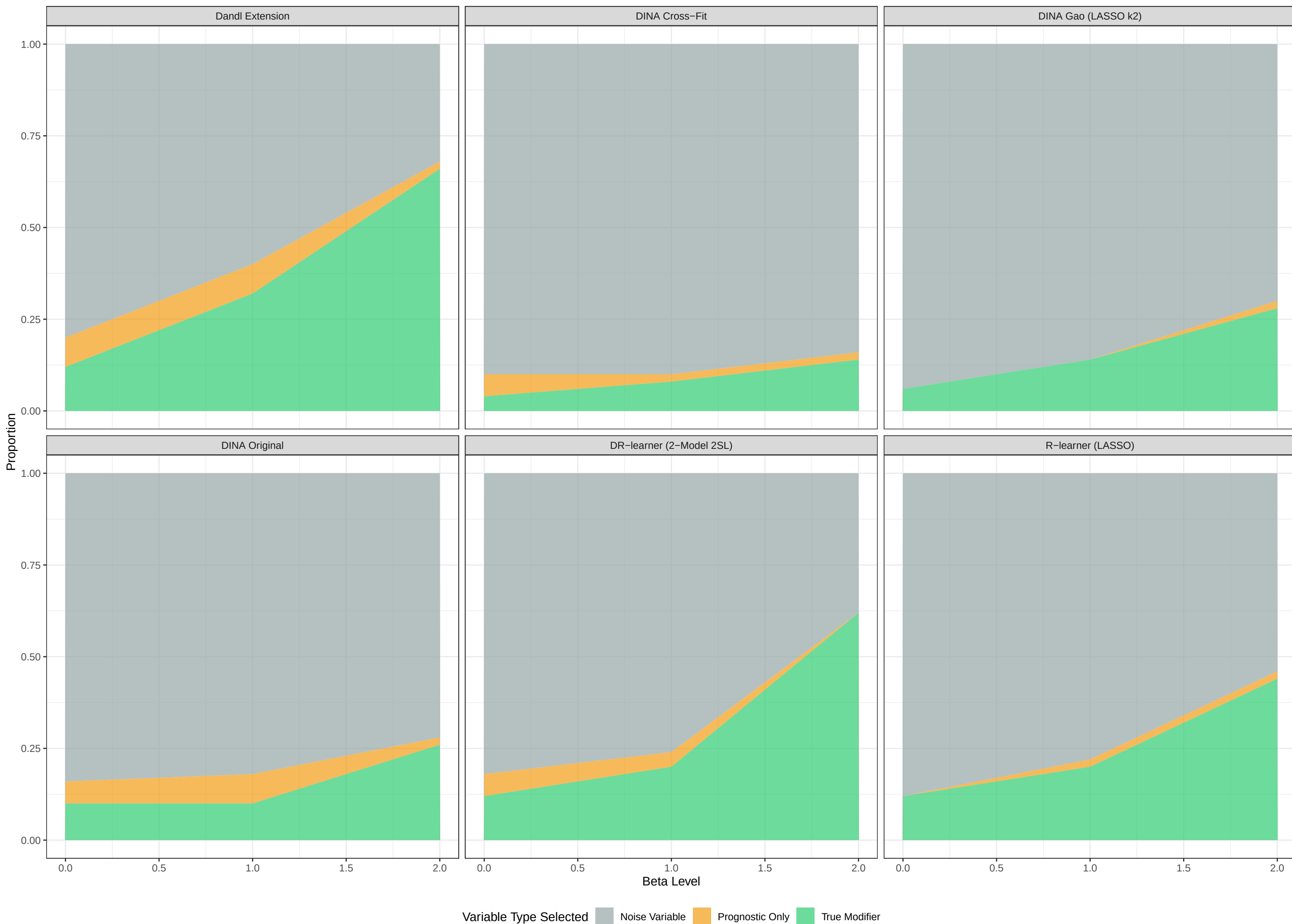
Objective 2(ii): Prob Top Variable is Truly Predictive (Scenario 4)

True modifiers: X14, X4



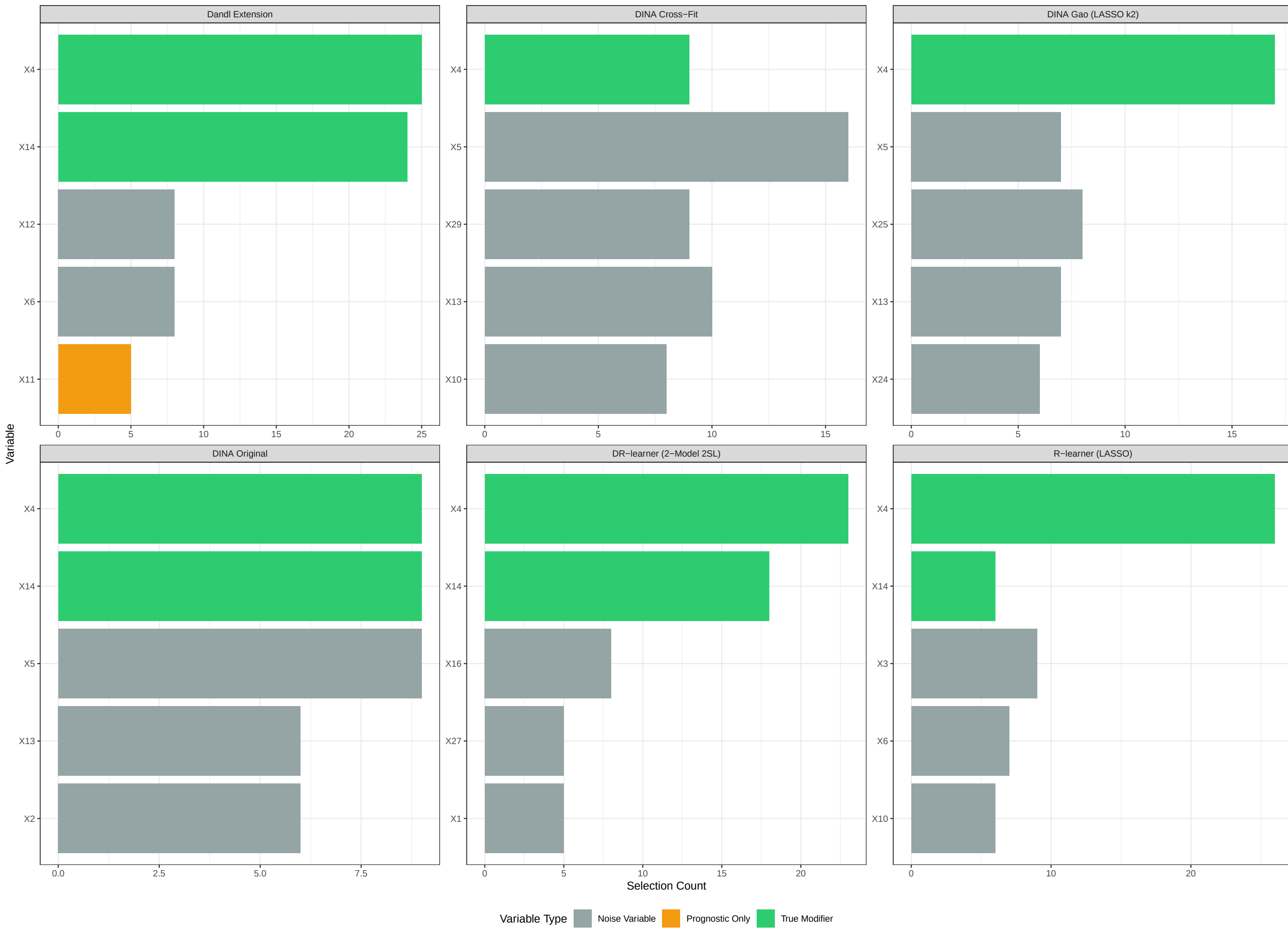
Objective 2(ii): Variable Type Selection Breakdown (Scenario 4)

Stacked area: what type of variable was selected as top modifier



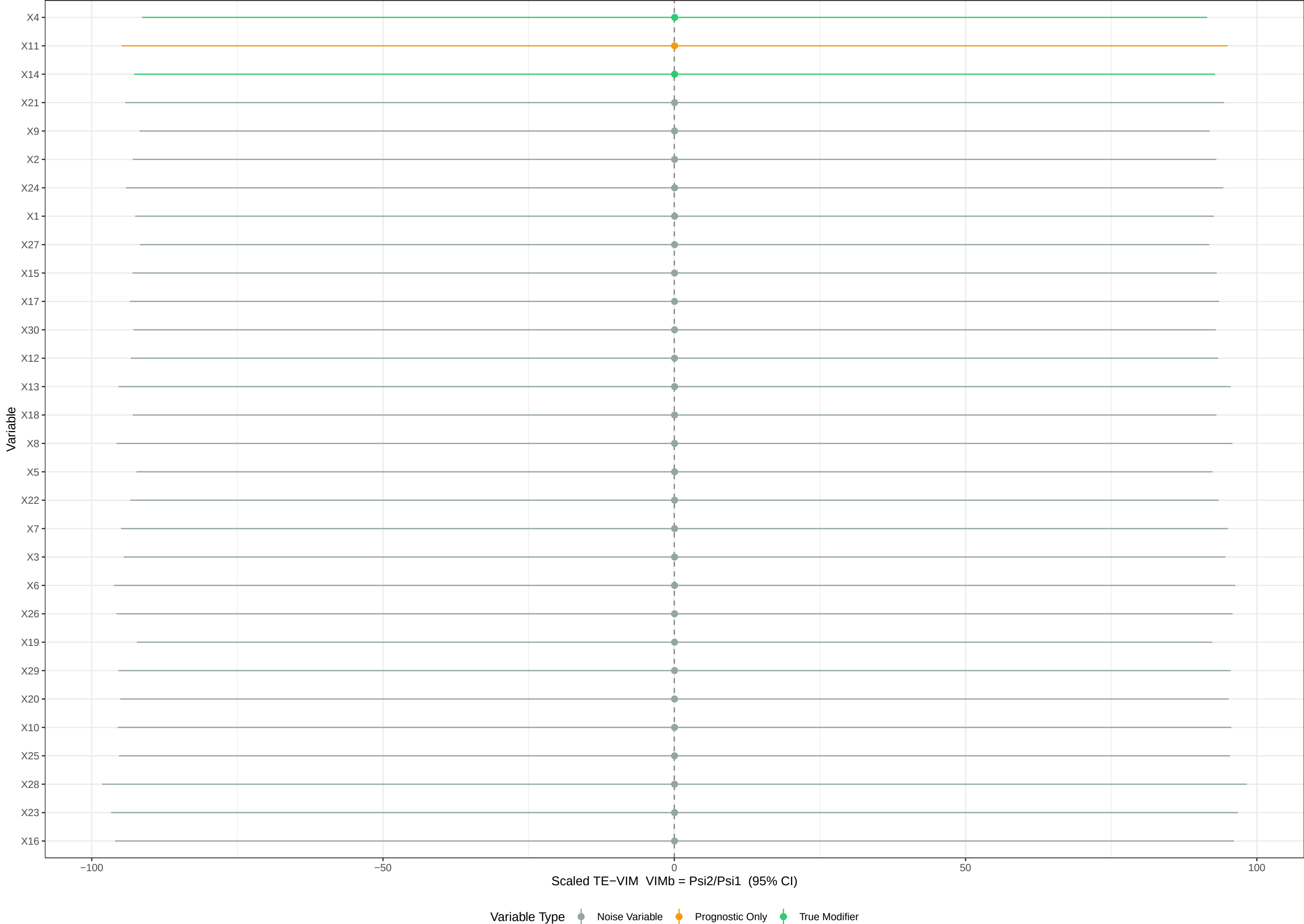
Objective 2: Most Frequently Selected Variables (Scenario 4, $\rho > 0$)

Top 5 variables selected most often across sims with heterogeneity

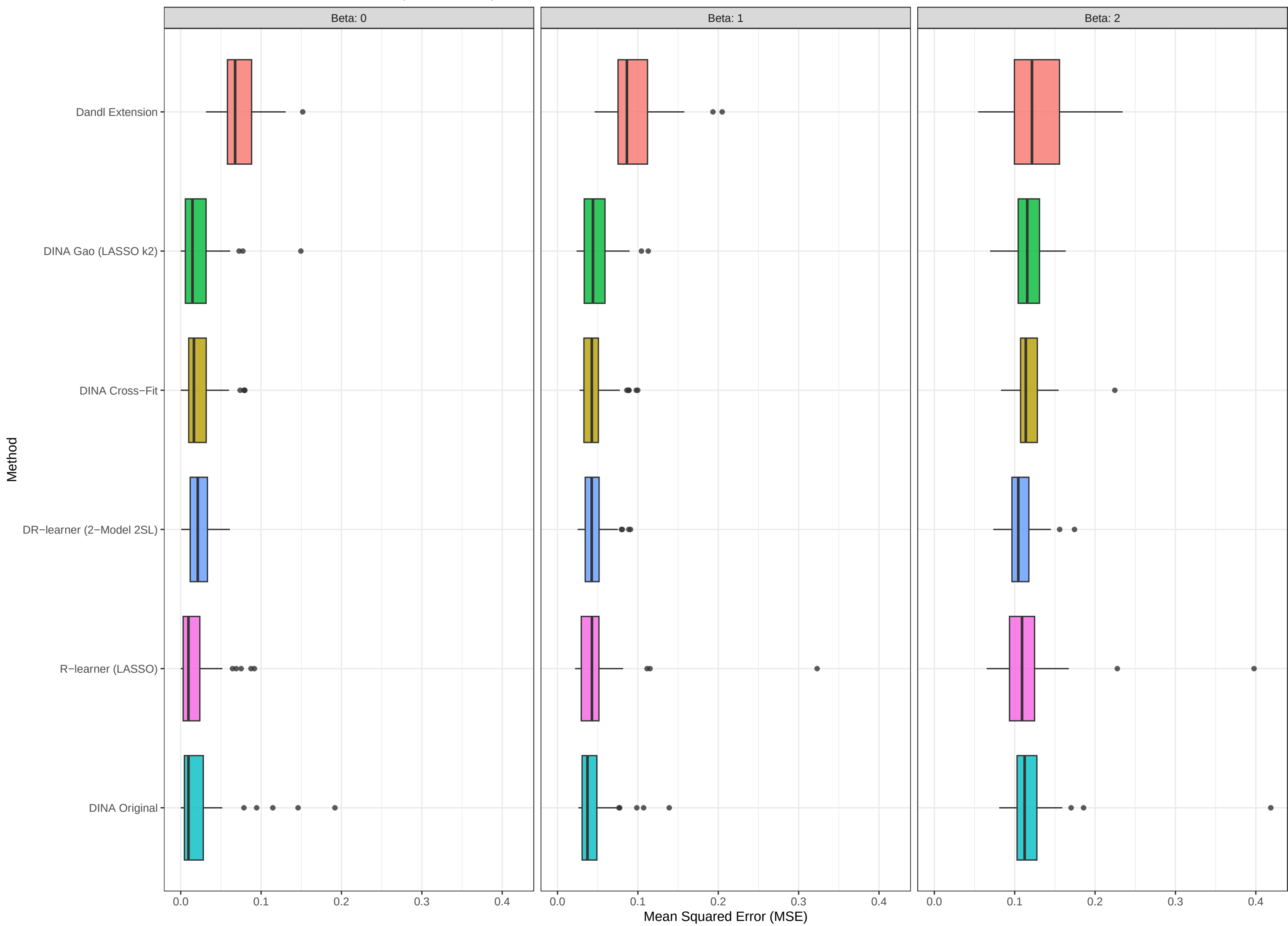


Objective 2 (TE-VIM TMLE): Scaled VIM Ranking (Scenario 4, beta>0)

Li 2026 VIMb = Psi2/Psi1 (TMLE), pooled over sims. True modifier(s): X14, X4

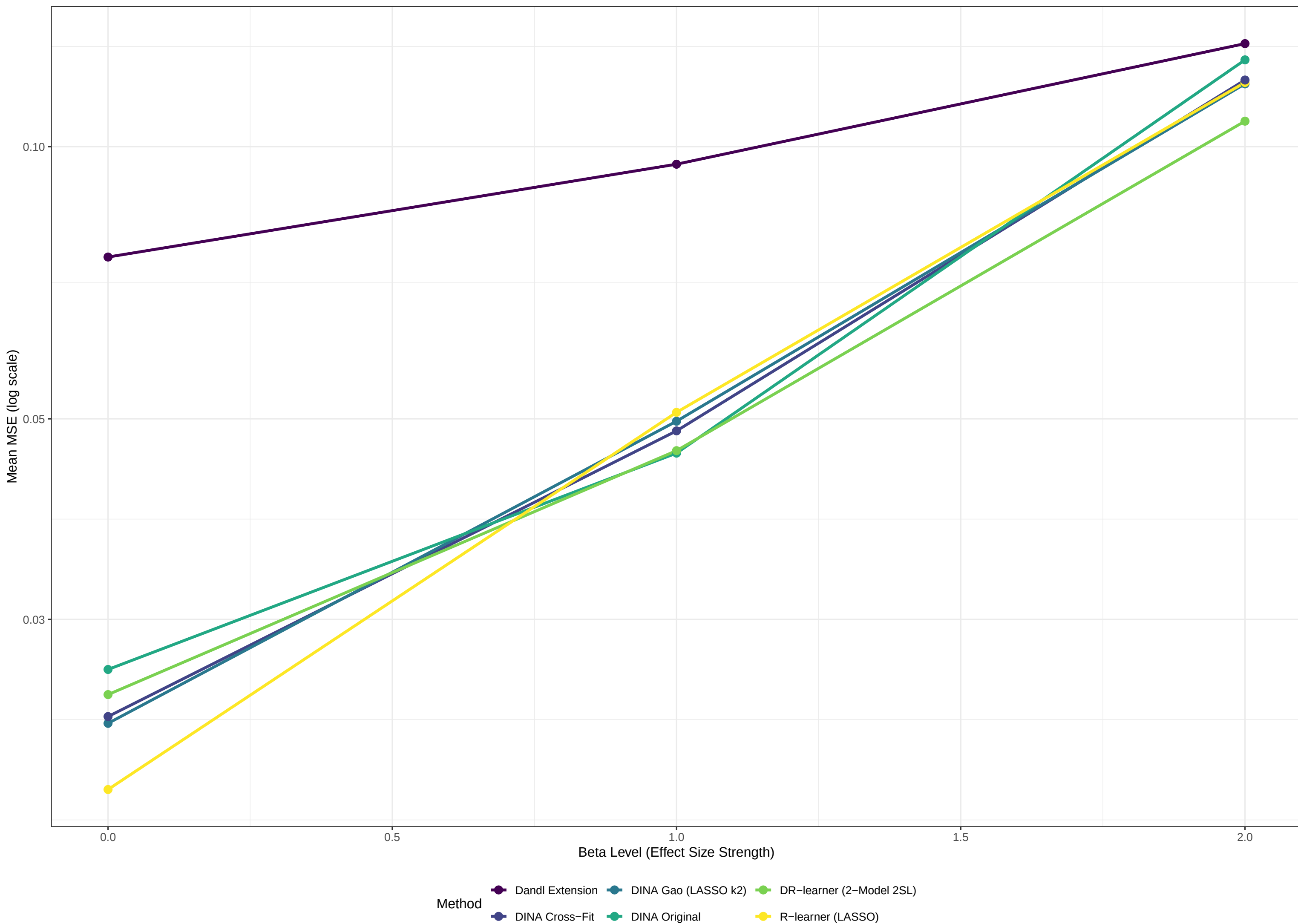


Objective 3: MSE Distribution (Scenario 4)
 Lower MSE = better CATE estimation. Spread = stability.

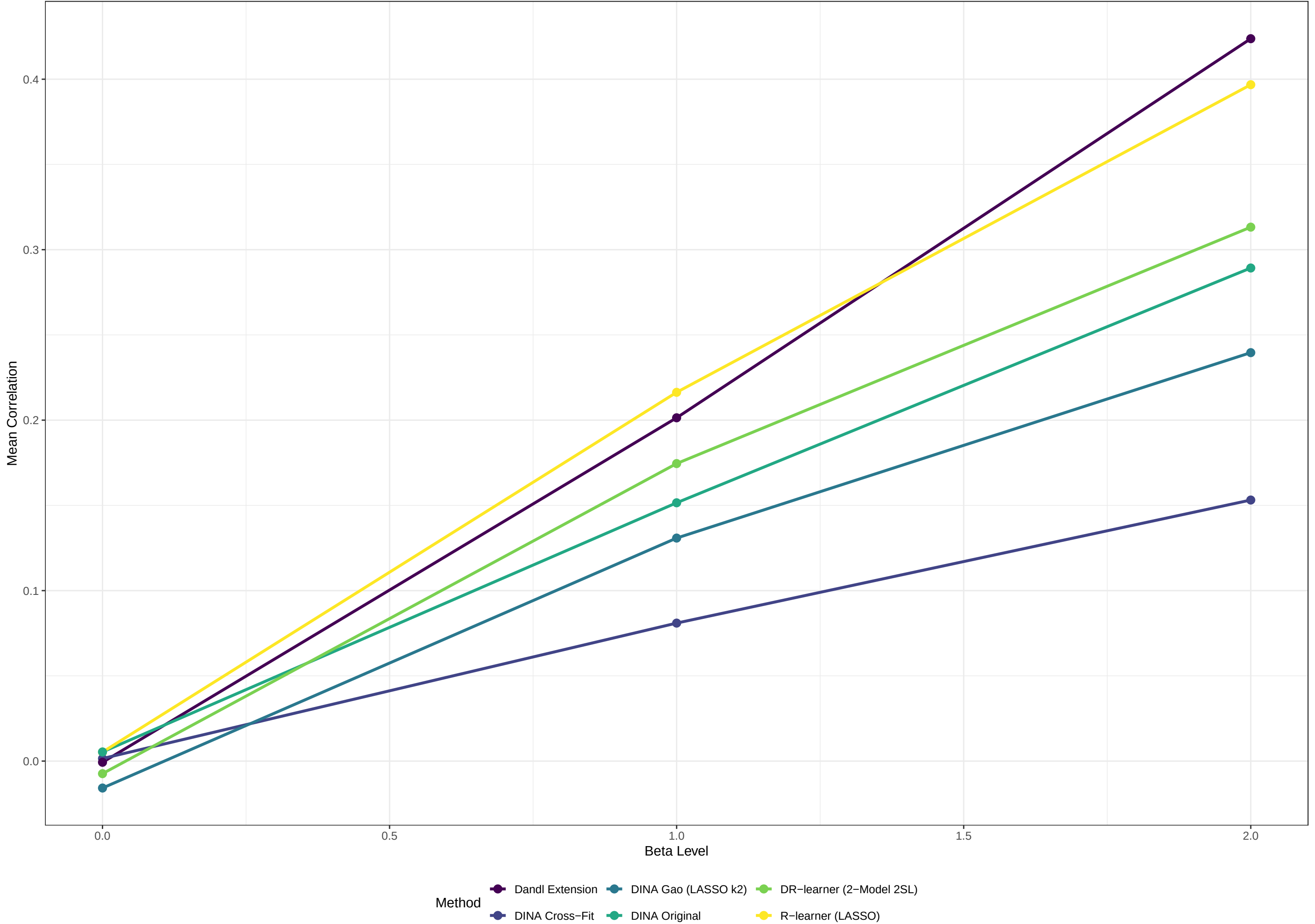


Objective 3: CATE Estimation Performance (Scenario 4)

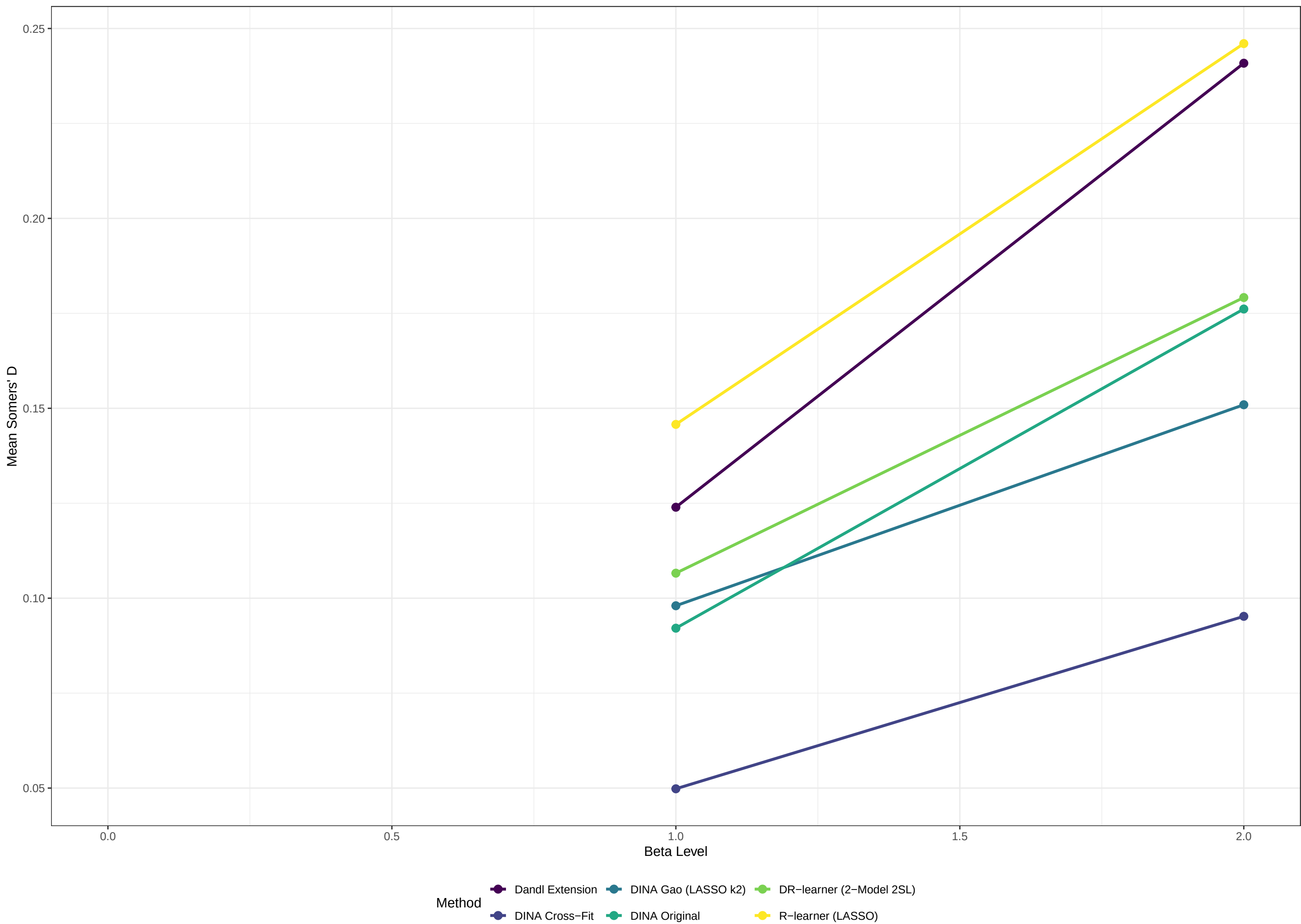
Mean Squared Error for estimating individual treatment effects



Objective 3: Correlation with True Effects (Scenario 4)
Higher correlation = better alignment with true treatment effects



Objective 3: Somers' D Rank Correlation (Scenario 4)
Sechidis et al. (2026) recommended Obj-3 metric; higher = better ranking



Summary Comparison (Scenario 4) – averaged over .>0
True Modifiers: X14, X4

Method	Avg_Power	Avg_Detection	Avg_MSE	Avg_Correlation	Avg_Somers_D	Avg_True_Selection
DR-learner (2-Model 2SL)	0.662	13	0.076	0.244	0.143	41
DINA Cross-Fit	0.660	15	0.084	0.117	0.073	11
Dandl Extension	0.660	15	0.113	0.313	0.182	49
DINA Original	0.660	16	0.085	0.220	0.134	18
R-learner (LASSO)	0.660	16	0.084	0.307	0.196	32
DINA Gao (LASSO k2)	0.658	17	0.084	0.185	0.124	21